

## Tutorial - Week 13

TG 11 - TG 12

1. Find the radius and the interval of convergence of the following series.

$$(a) \sum_{n=0}^{\infty} n x^n.$$

$$(b) \sum_{n=1}^{\infty} \frac{1}{n} (x-3)^n.$$

$$(c) \sum_{n=0}^{\infty} \frac{1}{n^2+1} (x-2)^n.$$

$$(d) \sum_{n=0}^{\infty} n! (2x-1)^n.$$

$$(e) \sum_{n=0}^{\infty} \frac{1}{n^3} (5x-4)^n.$$

$$(f) \sum_{n=0}^{\infty} \frac{(-3)^n}{\sqrt{n+1}} x^n.$$

$$(g) \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n.$$

$$(h) \sum_{n=2}^{\infty} \frac{1}{2^n \ln n} (x+2)^n.$$

$$(i) \sum_{n=1}^{\infty} \frac{\sqrt{n}}{9^n (x+7)^n}.$$

$$(j) \sum_{n=2}^{\infty} \frac{1}{n^n} (x-2)^n.$$

$$(k) \sum_{n=1}^{\infty} \frac{1}{5^n \sqrt{n}} (3x-1)^n.$$

$$(l) \sum_{n=1}^{\infty} \frac{n}{b^n} (x-a)^n, b > 0.$$

$$(m) \sum_{n=2}^{\infty} \frac{n2^n}{2 \cdot 4 \cdot 6 \cdots (2n)} x^n.$$

$$(n) \sum_{n=1}^{\infty} \frac{1}{n^3} (5x-4)^n.$$

$$(o) \sum_{n=1}^{\infty} \frac{1}{1 \cdot 3 \cdot 5 \cdots (2n-1)} x^n.$$

$$(p) \sum_{n=1}^{\infty} \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)} x^n.$$

2. Represent the given function  $f(x)$  as a power series and find its radius and the interval of convergence.

$$(a) f(x) = \frac{1}{1-x}.$$

$$(b) f(x) = \frac{x^5}{1-x}.$$

$$(c) f(x) = \frac{1}{(1-x)^2}.$$

$$(d) f(x) = \ln(x+1).$$

$$(e) f(x) = \arctan x.$$

$$(f) f(x) = \frac{x^2}{x^4+16}.$$

$$(g) f(x) = \frac{x}{2x^2+1}.$$

$$(h) f(x) = \frac{x-1}{x+2}.$$

$$(i) f(x) = \frac{x+a}{x^2+a^2}, a > 0.$$

$$(j) f(x) = \frac{2x-4}{x^2-4x+3}.$$

$$(k) f(x) = \frac{2x+3}{x^2+3x+2}.$$

$$(l) f(x) = \ln(5-x).$$

$$(m) f(x) = x^2 \arctan(x^3).$$

$$(n) f(x) = \left( \frac{x}{2-x} \right)^3.$$

$$(o) f(x) = \int \frac{x}{1-x^8} dx.$$

$$(p) f(x) = \int \frac{x}{1+x^3} dx.$$

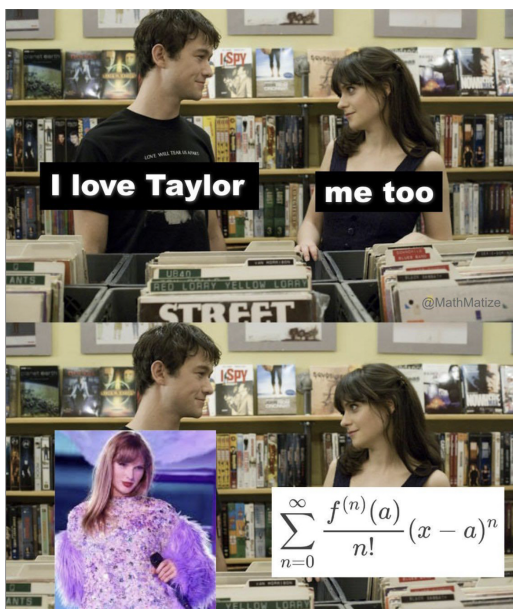
$$(q) f(x) = \int x^2 \ln(1+x) dx.$$

$$(r) f(x) = \int \frac{\arctan x}{x} dx.$$

3. Show that

$$\frac{1}{(1-x)^{m+1}} = \sum_{n=0}^{\infty} \binom{n+m}{m} x^n$$

for all  $m \in \mathbb{N}$  and all  $x \in (-1, 1)$  (*Hint: derivative of a power series*).



4. Find the Taylor series representation for

(a)  $\sin x$  at  $a = 0$ .

(b)  $\sin x$  at  $a = \pi$ .

(c)  $\sin^2 x = \frac{1}{2}(1 - \cos(2x))$  at  $a = 0$ .

(d)  $x \cos x$  at  $a = 0$ .

(e)  $\frac{1}{x+5}$  at  $a = -2$ .

(f)  $\frac{x^2 + 4x + 4}{x + 5}$  at  $a = -2$ .

(g)  $x^2 + 2x - 1$  at  $a = 1$ .

(h)  $(x^2 + 2x - 1)e^x$  at  $a = 1$ .

(i)  $e^{-x^2}$  at  $a = 0$ .

(j)  $\int e^{-x^2} dx$  at  $a = 0$ .

5. Find an approximation of  $f(x)$  with an error smaller than  $\varepsilon$ .

(a)  $f(x) = \sin(x)$ ,  $x = \frac{1}{2}$  and  $\varepsilon = 10^{-2}$ .

(b)  $f(x) = \int_0^x e^{-t^2} dt$ ,  $x = 1$  and  $\varepsilon = 10^{-3}$ .

